

KRISHNA KIREETI KASINADHUNI

Embedded Software/Firmware Engineer — ECE Final Year

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Summary

Final-year ECE student specializing in embedded firmware development. Hands-on experience in PWM, ADC, interrupts, sensor fusion, and PID control through UAV and visible light communication (VLC) systems, with strong fundamentals in real-time hardware–software integration.

Education

B.Tech – Electronics & Communication Engineering

2022 – 2026

Rajiv Gandhi University of Knowledge Technologies, Nuzvid

CGPA: 8.22

Technical Skills

Programming: C (Embedded), Python

Embedded Systems: Timers, Interrupts, ADC, PWM

Communication Protocols: UART, SPI, I2C

Microcontrollers: Arduino, ESP32, RaspberryPI

Firmware Concepts: Bare-metal programming, FreeRTOS (basic tasks & scheduling), Sensor Interfacing

Tools: Arduino IDE, Mission Planner, Multisim

Basic Exposure: Linux, KiCad, Machine Learning fundamentals

Experience

Research Intern – Embedded Systems

May 2025 – July 2025

IITDM Kurnool

RaberryPi, Quadcopter (Drone)

- Executed PWM-based LED modulation and ADC signal acquisition for VLC-based UAV localization in indoor environments.
- Applied PID tuning and IMU + altimeter sensor fusion to improve UAV stability and reduce drift during test flights.
- Optimized ESC timing and firmware integration to achieve smoother and more reliable flight control.

Projects

Li-Fi Based Optical Data Transmission System

Arduino

- Designed and implemented VLC transmitter and receiver using LED modulation and photodiode circuits.
- Developed firmware for data encoding and decoding, achieving ~75 characters per minute throughput.
- Tested system reliability under varying ambient light conditions.

RFID-Based Smart Guided Tourist Robot

Arduino

- Built an autonomous navigation robot using RFID-based indoor path tracking.
- Implemented microcontroller firmware for DC motor control and LCD-based navigation feedback.

Real-Time Multi-Sensor Flight Data Logger

STM32 & FreeRTOS

- Developed a FreeRTOS-based multi-sensor flight data logger integrating MPU-6050 IMU, BMP280 barometric sensor, and NEO-6M GPS on STM32F446RE.
- Implemented concurrent sensor acquisition, UART telemetry, and Python-based CSV logging for real-time flight and environmental data analysis.

Certifications

Microprocessors and Interfacing – NPTEL (IIT Guwahati)

Embedded for Beginners – NIELIT, Calicut

Understanding Incubation and Entrepreneurship – NPTEL (IIT Bombay)

Leadership

- Convenor, Teckzite'25:** Led organization of a large-scale technical fest and coordinated student teams.
- President, SDCAC:** Organized hackathons and mentored multiple student project teams.
- Additional Roles:** President, E-CRUSH — Promotions Manager, Teckzite — Volunteer, NSS & Student Mitra.